Martinez, Thomas - Education and Background Statement

I was born and raised in Miami, Florida, by Brazilian immigrant parents. I attended public schools, beginning in a Portuguese bilingual program and later joining a high school of more than 3,000 students. With most of my family living in Brazil, I found a second family in my school communities. The belonging I found there now motivates me to build community with and through mathematics. I want students with different backgrounds and educational experiences to have the support and opportunities to find their own place in that community.

I entered Harvey Mudd College intending to study computer science, but taking classes with Arthur Benjamin changed the direction of my education. His discrete mathematics course drew me toward combinatorics and sparked an interest that grew through further study, research, and conversations about interesting problems. As I became more involved in mathematics, I also became more connected to the people around it. Sharing questions and ideas gave me a way to form relationships through a subject I had come to love. I want my students to have opportunities to discover that same connection between intellectual curiosity and belonging.

After graduating in 2021, I returned to Miami as a City Year AmeriCorps member to support students returning to in-person learning. As Team Leader at Miami Jackson Senior High School, I taught four Algebra I classes for students at risk of not graduating on time. More than half had recently immigrated from countries including Honduras, El Salvador, and Cuba, and I taught them mathematics in Spanish. During the first week, many told me that math was their least favorite subject. Building relationships with these students became central to my teaching, alongside designing weekly interventions for those needing additional support. These connections were especially important for immigrant students, helping them feel more integrated into the school community. As the year progressed, students began to grasp material they had struggled with, and some told me they had started looking forward to class with me. Hearing this helped me understand how much a caring instructor can influence a student's experience of mathematics and opportunities to learn. It also made educational equity a personal responsibility: attending to students' language, educational circumstances, and sense of belonging was part of helping them learn.

At UCLA, undergraduate mentoring has given me a way to share the curiosity and connection I first found in Arthur Benjamin's classes. Through the Directed Reading Program, I have mentored eight undergraduates on topics ranging from graph theory to tiling proofs of Fibonacci identities. With Olha Shevchenko and three of my undergraduate mentees, I recently submitted a paper on q-Fibonacci combinatorics to a journal. I value this work as an opportunity to help students see themselves as participants in mathematics, with questions and ideas worth pursuing. I want research to offer them both an intellectual challenge and a community in which their contributions matter.

Tutoring first- and second-year students through Harvey Mudd's Core curriculum gave me an early opportunity to help students support one another as they adjusted to college mathematics. At UCLA, working with undergraduate Learning Assistants in my discussion sections extended this approach through active learning. In these settings, I want students to feel comfortable asking questions, explaining their thinking, and working through uncertainty together. As mathematics LA coordinator, I connected instructors with the program and helped extend it to additional courses. My goal is for students with different levels of preparation and confidence to find support from their peers and feel that they belong in mathematics from their first college courses.

As a postdoctoral fellow at UC Davis, I would seek to mentor undergraduates through the Davis Math Lab, developing combinatorics projects with accessible starting points and room for sustained investigation. As a faculty member, I hope to establish a summer Research Experiences for Undergraduates program involving graduate students and faculty as mentors, while continuing to support peer instruction. Especially in this age of AI, I want to emphasize the human side of mathematics and build community. I hope to offer students what my own educational communities offered me: a place to grow, contribute, and belong.